

Affordable, Portable, Real-Time Chlorophyll Sensor

Physics Report | Group 04

Group 04

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Executive Summary

This report details the development of a portable, high-accuracy chlorophyll sensor utilizing **Laser-Induced Fluorescence (LIF)** and **tinyML** algorithms. Designed for agricultural, herbal, and aquatic monitoring, the system provides real-time, calibrated measurements of both Chlorophyll-a (Chl-a) and Chlorophyll-b (Chl-b) concentrations.

1 Introduction

1.1 Importance of Chlorophyll

Chlorophyll is a primary indicator of plant health, product quality in the herbal industry, and algal biomass in aquatic ecosystems. Monitoring its concentration is critical for water quality

assessment and precision agriculture.

1.2 Detection Methodology

While traditional methods (HPLC, Mass Spectrometry) offer high precision, they are expensive and laboratory-bound. Existing affordable meters often provide only relative content and lack absolute calibration.

- **Research Gap:** Creating an affordable sensor providing calibrated, simultaneous measurements of Chl-a and Chl-b.
- **Objective:** Prototype a real-time sensor using LIF and on-device edge computing analysis.

2 Theoretical Framework

2.1 Physics of Laser-Induced Fluorescence (LIF)

The sensor employs a flow-through system where samples are excited by targeted wavelengths: * **Laser 405 nm:** Optimized for Chl-a excitation. * **Laser 450 nm:** Optimized for Chl-b excitation.

2.2 Spectral Red Shift & Concentration Features

A critical physical observation in this study is the **Red Shift** of the fluorescence peak as concentration increases. This phenomenon occurs due to: * **Inner Filter Effect (IFE):** Re-absorption of the shorter-wavelength side of the emission spectrum by the chlorophyll matrix. * **Self-Absorption:** The overlap between chlorophyll absorption and emission bands causes selective re-absorption at shorter wavelengths. * **Concentration Quenching:** At high concentrations, molecular aggregation further shifts the emission to longer wavelengths.

These effects produce three measurable spectral features that vary systematically with concentration, as confirmed by our simulation study:

Unlike static peak-height models, our PLS and SVR models analyze the **entire spectral curve**, allowing them to correlate this non-linear shift with absolute concentration. Importantly, our simulation study confirmed that **retaining the red shift** (i.e., not aligning peaks) provides superior model performance compared to peak-aligned spectra.

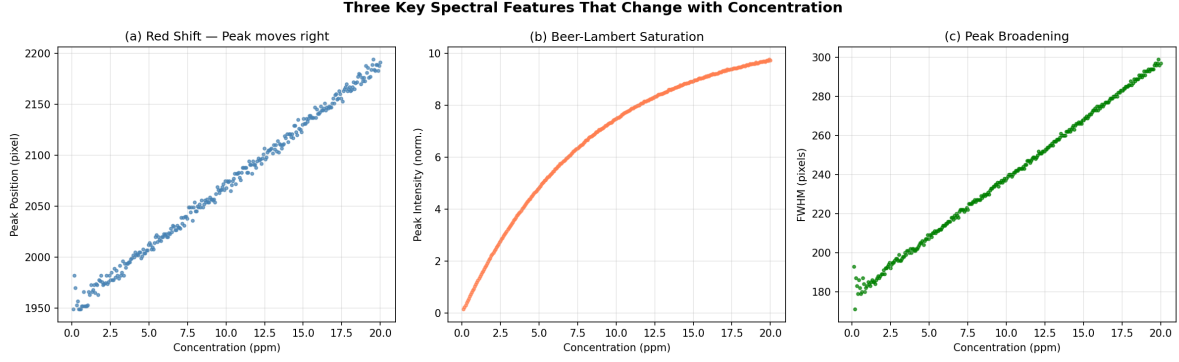


Figure 1: Three key spectral features that change with chlorophyll concentration: (a) peak position shifts ~ 240 pixels across the 0–20 ppm range, (b) intensity saturates following Beer-Lambert law, and (c) FWHM broadens linearly.

3 Experimental Setup & Hardware

3.1 System Architecture

- **Control Unit:** STM32H743VIT6 (High-performance ARM Cortex-M7 running at 480MHz) for high-speed acquisition and on-device tinyML inference.
- **Sensor:** Toshiba TCD1304 linear CCD with 3648 active pixels, providing high spatial resolution for fluorescence spectroscopy.
- **Optical Path:**
 - **Excitation:** Dual-laser system (405nm and 450nm) targeting the Soret bands of Chl-a and Chl-b.
 - **Dispersion:** 1200 lines/mm holographic diffraction grating to resolve the fluorescence spectrum.
 - **Collimation:** Objective lenses used for beam collimation to ensure a sharp spectral image on the CCD.
- **PCB Design:** A custom PCB was developed using **KiCad** and manufactured via **JLCPCB** to ensure signal integrity and portability.

3.2 System Connectivity

The following diagram illustrates the physical and logical connections between the core components:

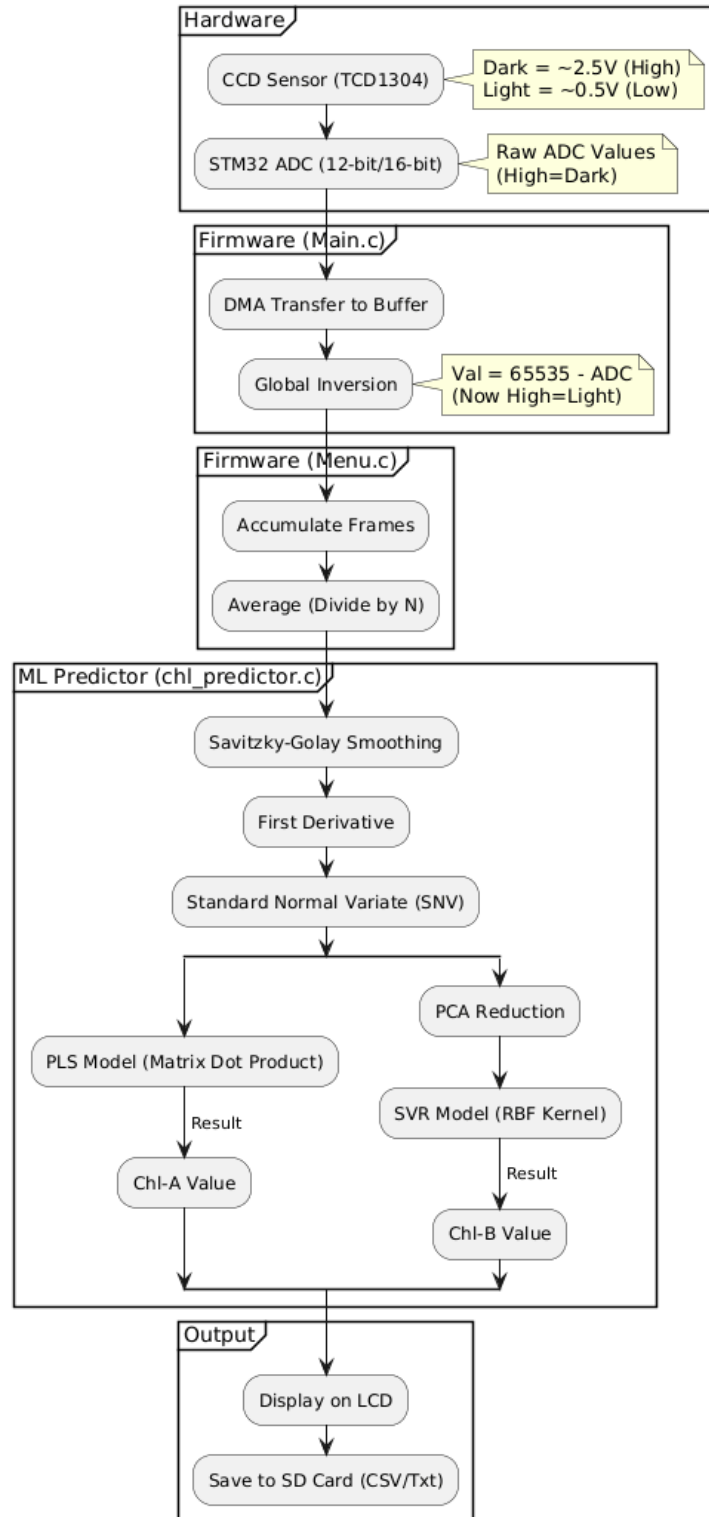


Figure 2: System connectivity diagram showing physical and logical connections between STM32H743VIT6, TCD1304 CCD, lasers, servo, LCD, SD card, and USB

3.3 Information Pipeline

The data flow from raw photons to on-device concentration prediction is summarized below:

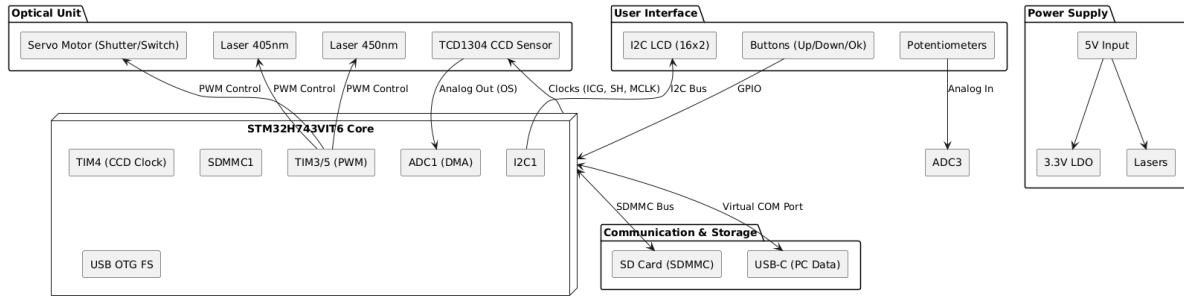


Figure 3: Software architecture class diagram showing the modular firmware design with Core Application, Signal Processing, and Storage packages

3.4 Circuitry and PCB Design

The system's electronic backbone is a custom-designed PCB optimized for low-noise CCD readout and high-speed signal processing.

Note: The full vector PCB schematic is available in the electronic HTML version of this report. Refer to Figure 4 for the physical board implementation.

3.4.1 Key Hardware Features

- **MCU High-Speed Interfacing:** The STM32H7 communicates with the TCD1304 CCD using high-speed timers to generate the Master Clock (fM) and Shift Gate (SH) pulses, ensuring synchronized pixel acquisition.
- **Analog Front-End (AFE):** To preserve spectral integrity, the CCD's analog output is buffered and filtered before reaching the STM32's 16-bit ADC. This minimizes quantization noise and electronic interference.
- **Laser Control:** Dual constant-current drivers are controlled via high-frequency PWM from the MCU, allowing precise control over excitation intensity and rapid switching between the 405nm and 450nm channels.
- **Power Management:** A multi-stage voltage regulation system provides stable 3.3V and 5.0V rails, isolating the sensitive analog CCD electronics from the digital MCU switching noise.

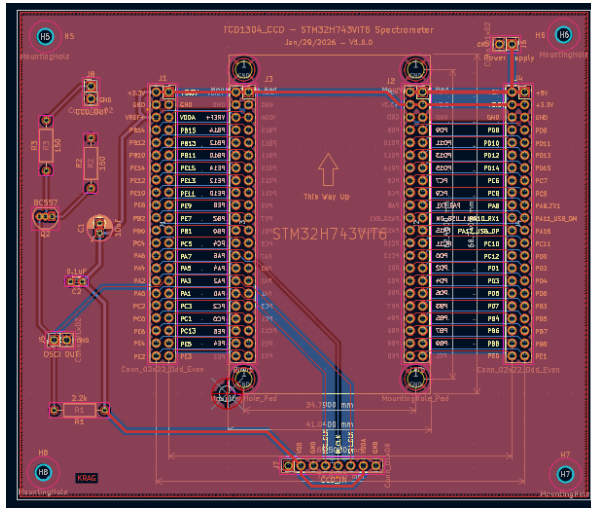


Figure 4: KiCad PCB Layout Implementation

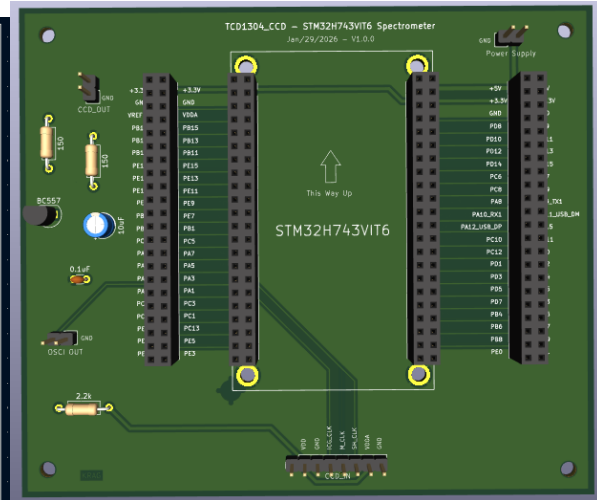


Figure 5: PCB 3D Visualization (Front Perspective)

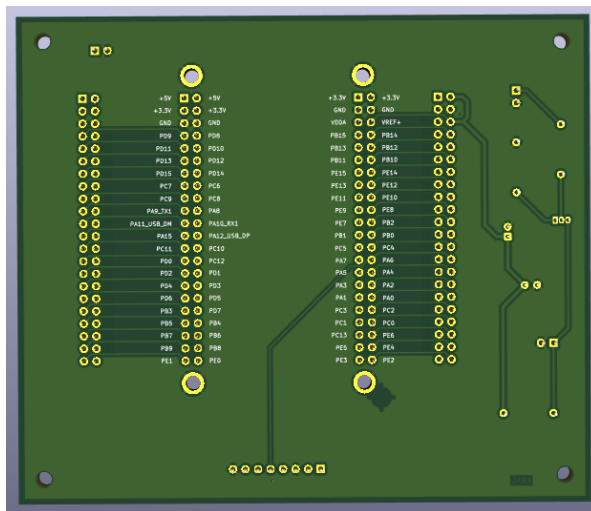


Figure 6: PCB 3D Visualization (Rear Connectivity)

3.5 Project Economics (Budget)

The system is designed for maximum affordability without compromising spectral precision. *

Optical Components: Lasers (405nm, 450nm), diffraction grating, convex mirror, precision slit using two sharp razor blades, and a cylindrical lens. * **Electronics:** STM32H743VIT6 MCU, TCD1304 CCD, 16x2 LCD Module with I2C interface, two 10k Ω potentiometers, two 2N2222 transistors, two 4.7k Ω resistors, two 1k Ω resistors, and one servo motor. * **Manufacturing:** Custom PCB (KiCad/JLCPCB), and a custom 3D-printed cuvette holder. * **Infrastructure:** 3D-printed housing and a flow-through sample chamber.

4 Software & Automation

4.1 Firmware & Software Design

The system's firmware is modularized to handle high-speed data acquisition, real-time UI updates, and machine learning inference concurrently.

4.2 Automated Operational Logic

To ensure repeatable measurements and optimal data quality, the system implements automated state machines for measurement and exposure tuning.

4.2.1 Auto-Measurement Sequence

The following state machine describes the automated dual-laser measurement cycle:

4.2.2 Auto-Exposure Optimization

The system utilizes a proportional P-control logic to automatically tune the CCD integration time, maximizing signal-to-noise ratio without saturating the sensor pixels.

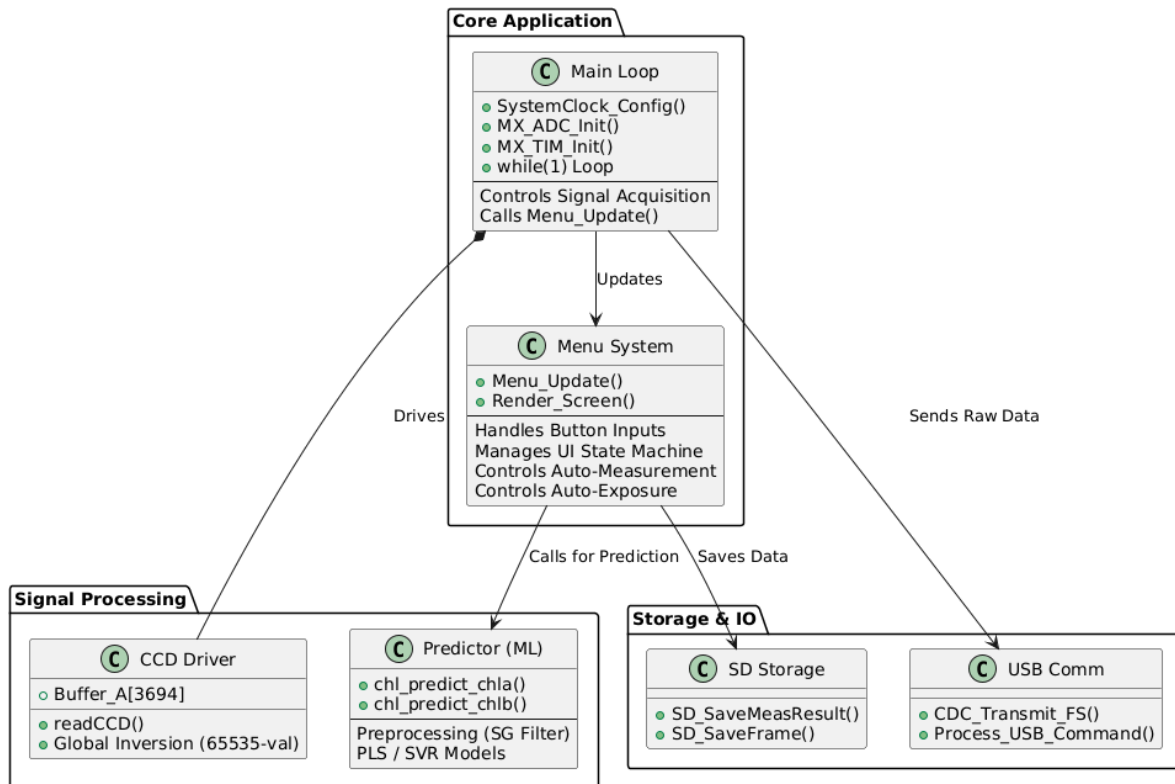


Figure 7: State machine diagram for the automated dual-laser measurement sequence

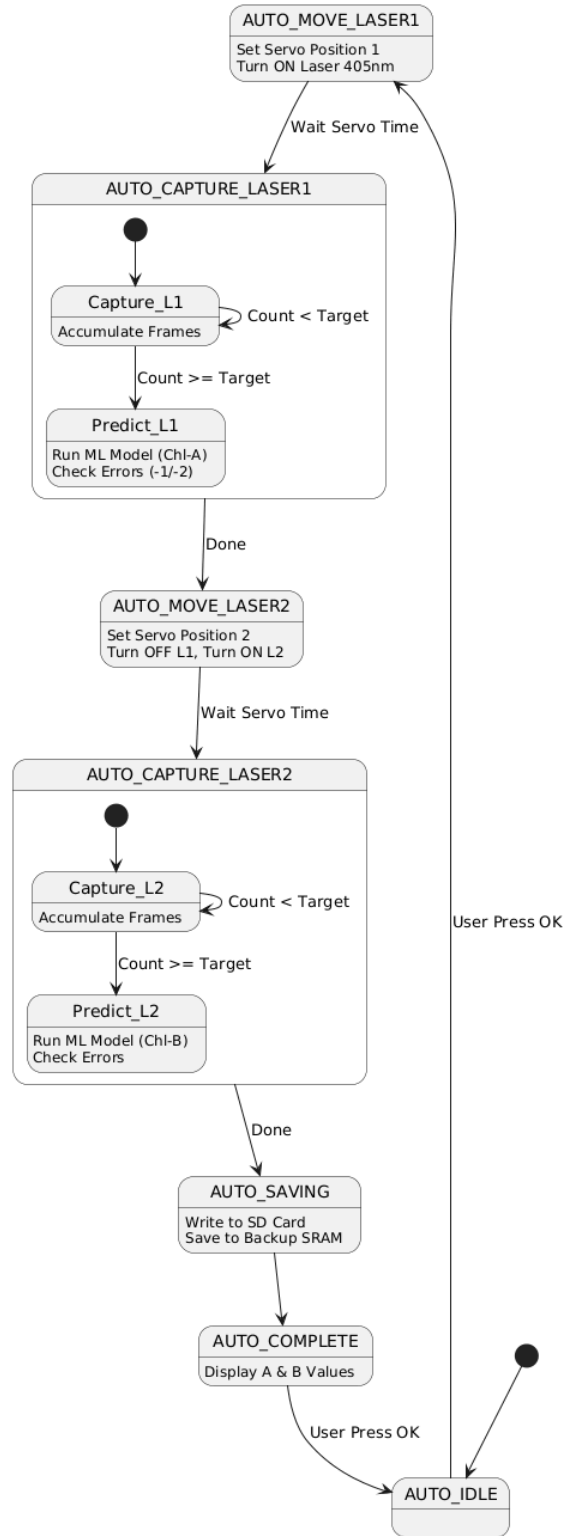


Figure 8: State machine diagram for the auto-exposure P-control optimization loop

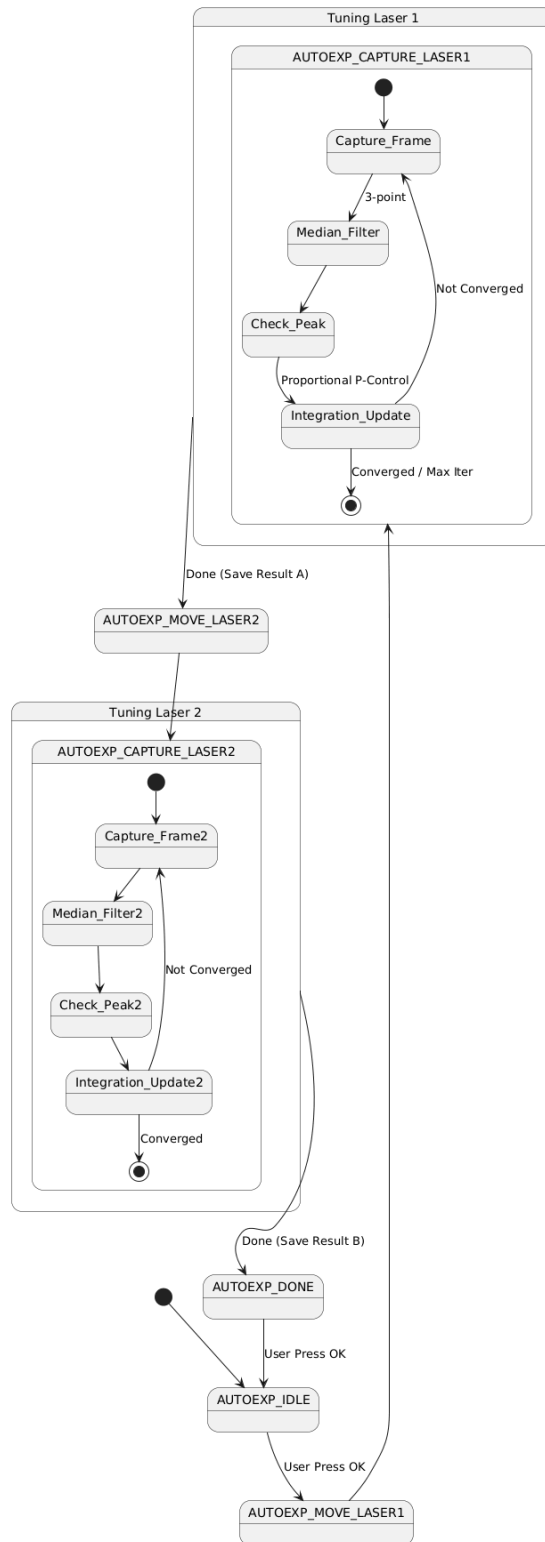


Figure 9: Additional system diagram

5 Methodology: Signal Processing & ML

5.1 Digital Signal Processing Pipeline

To mitigate noise and baseline drift, the raw CCD data (3648 pixels) undergoes a rigorous cleaning pipeline before inference:

$$\text{Clean Signal} = (65535 - \text{ADC Value}) - \text{Dark Signal}$$

1. **Inversion & Dark Subtraction:** CCD output is inverted ($\text{Voltage} \propto 1/\text{Intensity}$), and a reference dark spectrum is subtracted to remove fixed-pattern noise.
2. **Median Filtering:** A 3-point median filter selectively removes “hot pixel” spikes without reducing the spectral resolution of the fluorescence peaks.
3. **Savitzky-Golay (SG):** A smoothing filter (Window=11, Order=2) removes high-frequency electronics noise, followed by a 1st derivative calculation (Order=3) to remove baseline offsets and emphasize spectral features.
4. **Standard Normal Variate (SNV) & Normalization:** The signal is normalized by its mean and standard deviation:

$$SNV(x) = \frac{x - \mu}{\sigma}$$

Furthermore, intensity is normalized by integration time (ms), making the model **exposure-independent**.

5.2 Edge Machine Learning Implementation

Built with **Edge Impulse**, the models run directly on the STM32: * **Chl-a:** Partial Least Squares (PLS) Regression. * **Chl-b:** Support Vector Regression (SVR) with RBF Kernel.

A simulation-based comparison between PLS and a **1D Convolutional Neural Network (CNN)** was conducted to evaluate future algorithm improvements (see Section 6).

5.3 Case Study: Model Development & Iteration

The gap between training R^2 (0.99) and cross-validated test R^2 (0.85–0.87) is a classic indicator of overfitting. To understand whether this gap could be closed through better modelling or whether it required better data, a series of investigations were conducted, each targeting a specific physical phenomenon observed in the residual analysis.

5.3.1 Issue 1 — Heteroscedastic Residuals (Non-Constant Variance)

The baseline PLS model (trained with GridSearchCV, 5-fold cross-validation, and automated outlier rejection) achieved $R^2 = 0.87$ with RMSE 1.55 ppm. However, the residual analysis (Figure 11) revealed a distinctive **funnel-shaped pattern** — prediction errors grew proportionally with concentration.

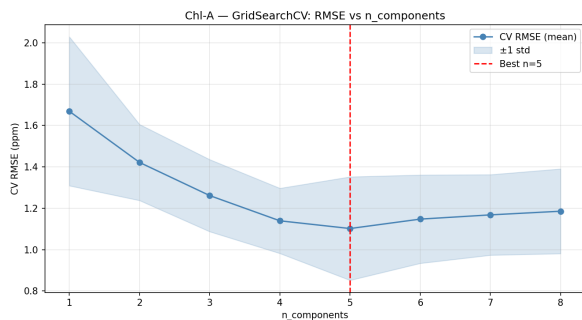


Figure 10: Baseline model: GridSearchCV selected 9 PLS components. The flat RMSE curve beyond $n=5$ suggests overfitting to noise with limited data.

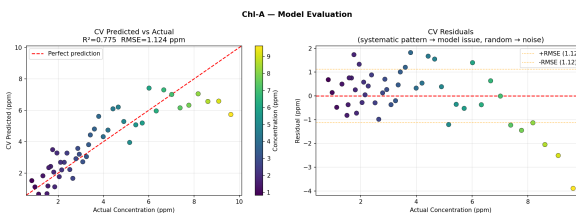


Figure 11: Baseline model: CV predicted vs actual (left) and residuals (right). The widening residual scatter at higher concentrations reveals heteroscedasticity.

Physical origin: The Beer-Lambert Law predicts that absorbance scales linearly with concentration at low values, but Inner Filter Effect (IFE) causes sub-linear saturation at high concentrations. This means the same ppm change produces a *smaller* spectral change at 15 ppm than at 2 ppm, leading to proportionally larger errors at higher concentrations.

Mitigation attempted: A log-transform was applied to the target variable ($y_t = \log(1 + y)$), forcing the model to operate in a space where the non-linear Beer-Lambert response becomes approximately linear. Figure 12 shows that the transformed residuals (right panel) exhibit **uniform variance** — confirming the heteroscedasticity was addressed.

Outcome: While the log-transform successfully equalised variance across the concentration range, the overall RMSE improved only marginally ($\sim 1.55 \rightarrow \sim 1.45$ ppm). This confirmed that the heteroscedasticity was a secondary effect — the primary limitation was **insufficient training data** to capture the full spectral diversity.

5.3.2 Issue 2 — Red Shift as a Concentration Feature

As documented in Section 2.2, the fluorescence peak position shifts rightward with increasing concentration due to self-absorption effects. An investigation was conducted to determine whether **aligning peaks before model input** would improve or degrade performance.

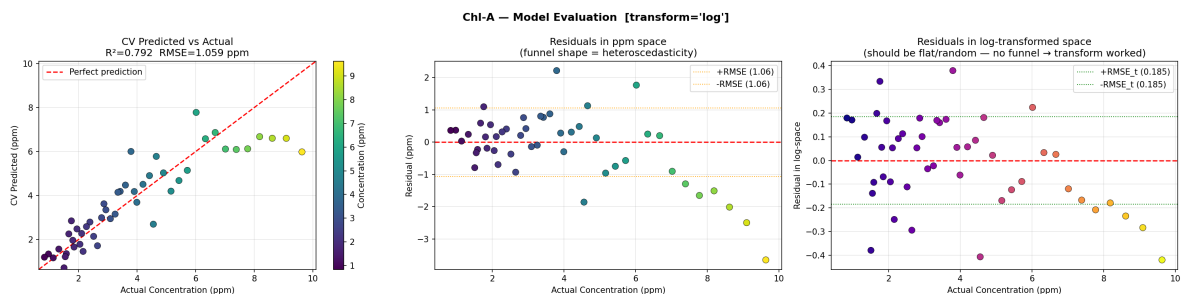


Figure 12: Log-transformed model evaluation. Left: predicted vs actual. Centre: residuals in ppm space still show spread. Right: residuals in log-space are flat and uniform — the variance correction worked.

Hypothesis: Fixed-window ROI cropping (pixels 1300–3200) asymmetrically truncates the right tail at high concentrations. Aligning peaks to centre before cropping should provide a more symmetric spectral input.

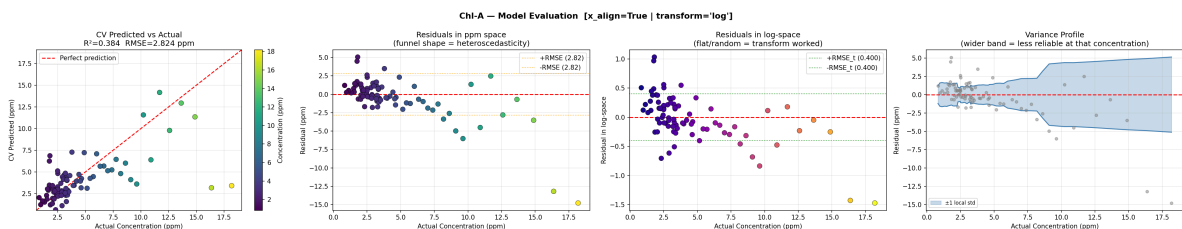


Figure 13: Model with peak alignment: Despite the theoretical advantage, performance degraded (R^2 0.85, $RMSE$ 1.60 ppm). Aligning peaks *removes* the red shift information that PLS uses as a concentration feature.

Outcome: Performance was **worse** with alignment (R^2 dropped from 0.87 to ~ 0.85). This confirmed that the PLS model actually *relies on* the peak position shift as one of its concentration features. Removing this information by alignment was counterproductive. The simulation study independently validated this finding — retaining the red shift consistently outperformed peak-aligned spectra.

5.3.3 Issue 3 — IFE Non-Linearity Across the Concentration Range

The concentration-dependent non-linearity from IFE suggested that a single linear model might struggle across the full 0–20 ppm range. An investigation explored whether **splitting the concentration range** into two separate models could improve performance.

Approach: Two independent PLS models were trained — one for concentrations ≤ 5 ppm (Beer-Lambert linear regime) and one for > 5 ppm (IFE-dominated regime). A sigmoid blending

function merged predictions smoothly at the boundary.

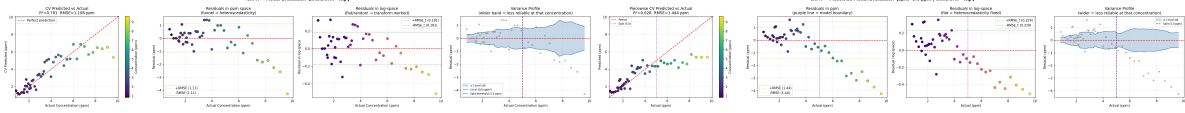


Figure 14: Single PLS model with variance pro-File (rightmost panel). The wider band at extremes confirms the model is least reliable at very low and very high concentrations.

Figure 15: Piecewise (split) model. The purple line marks the 5 ppm split threshold. Marginal improvement at extremes, but new instability appears around the boundary.

Outcome: With only ~32 Chl-a samples split into two groups (~15 each), each sub-model had **far too few training samples** to generalise. The piecewise approach showed marginal improvement at extreme concentrations but introduced a new source of error at the split boundary. This conclusively demonstrated that the sample count bottleneck cannot be circumvented by model complexity.

6 Comprehensive Simulation Study

To address the overfitting and limited data issues, a comprehensive **simulation study** was conducted using a physics-based synthetic spectrum generator that models the TCD1304 CCD's response to chlorophyll fluorescence. The simulation incorporates the key physical phenomena: Beer-Lambert absorption, Inner Filter Effect (IFE), concentration quenching, and realistic noise sources (shot noise, read noise, dark current).

The study quantified the effect of six key data collection parameters on PLS model accuracy:

6.1 Effect of Sample Count and Peak Alignment

The simulation confirmed that **200+ unique concentration samples** are required to achieve an RMSE below 0.3 ppm. Critically, **unaligned spectra** (preserving the red shift) consistently outperformed peak-aligned spectra at every sample count, demonstrating that the red shift is a valuable concentration feature.

6.2 Effect of Frame Averaging (Noise Reduction)

Averaging multiple CCD frames reduces noise proportionally to $\sqrt{N}$. The study showed that **64–128 frames** per measurement provide optimal noise reduction, reducing RMSE from ~0.9

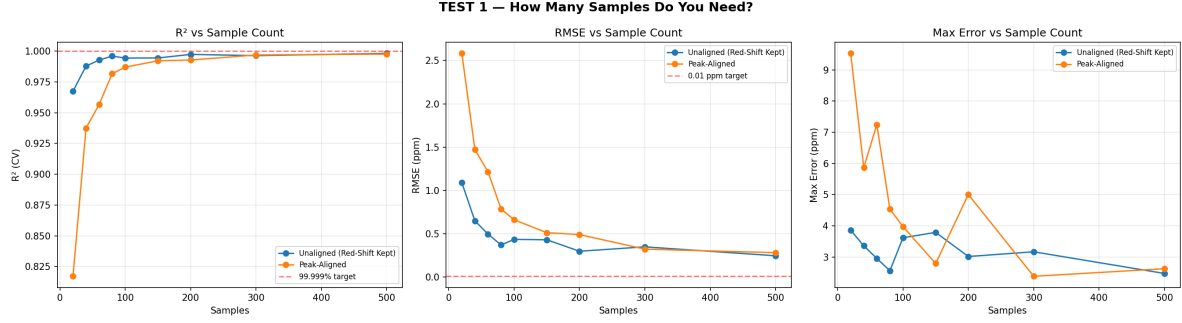


Figure 16: Effect of sample count on RMSE and R^2 for aligned vs. unaligned spectra. Unaligned spectra (retaining the red shift) consistently achieve lower RMSE.

ppm (single frame) to ~0.2 ppm (128 frames). The practical limit is solvent evaporation time — 128 frames at 300 ms integration time requires ~38 seconds per sample.

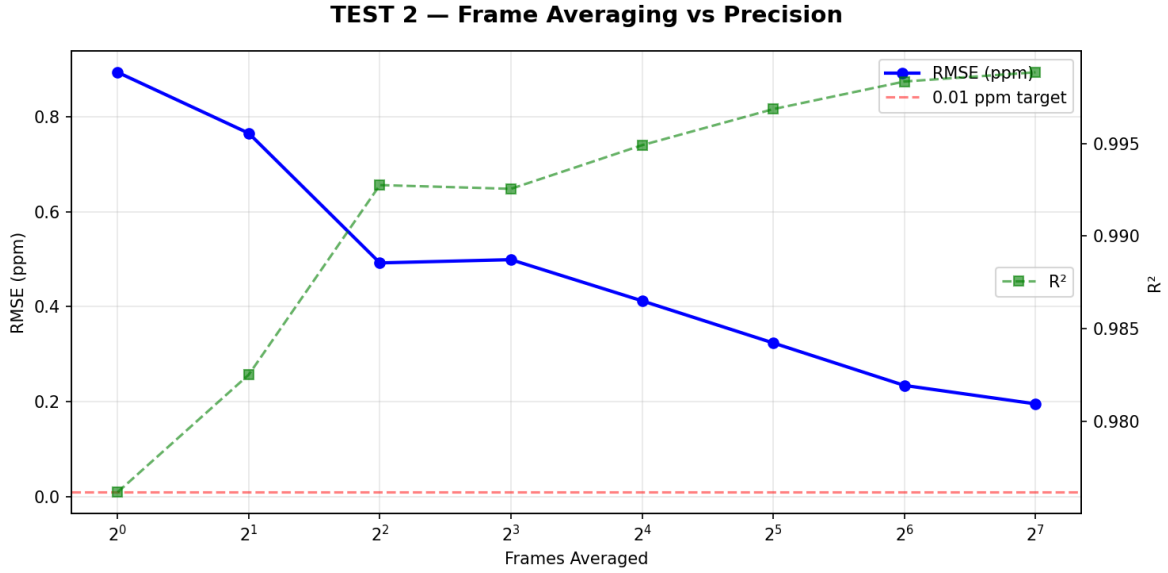


Figure 17: Effect of frame averaging on RMSE and SNR. The RMSE decreases proportionally to $1/\sqrt{N}$ as expected from noise theory.

6.3 Integration Time Optimization

The optimal integration time was found to be **300–700 ms**. Below 200 ms, the signal-to-noise ratio is insufficient. Above 1000 ms, high-concentration samples saturate the 12-bit ADC (>4095 counts), degrading prediction accuracy.

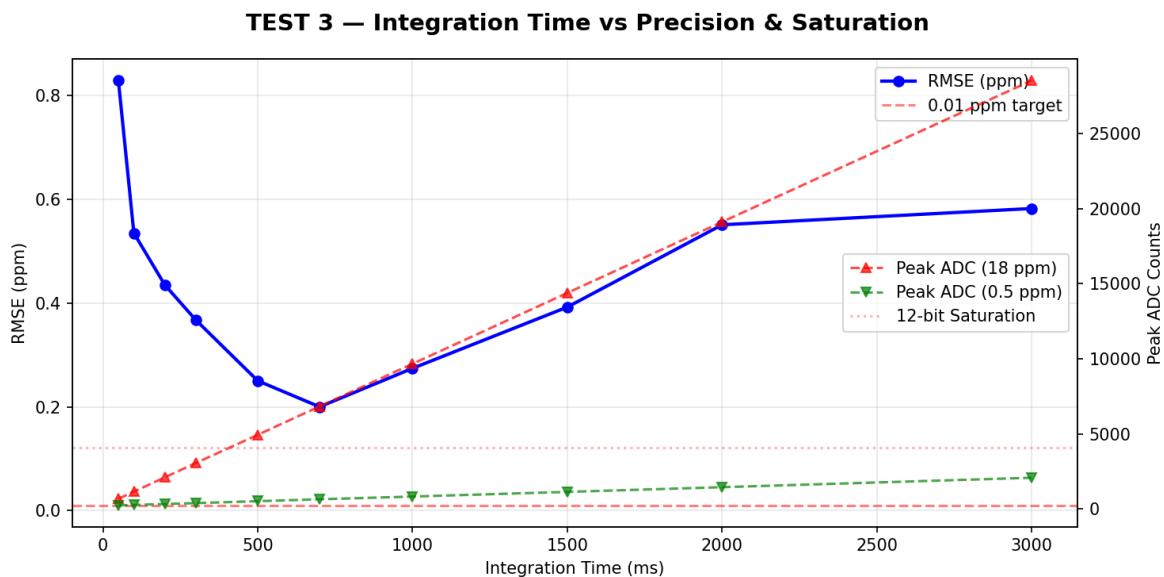


Figure 18: Integration time vs. RMSE and peak ADC counts. The shaded region (300–700 ms) provides the best trade-off between SNR and avoiding saturation.

6.4 1D CNN vs PLS Benchmarking

A 1D Convolutional Neural Network was prototyped and compared against PLS using a grid search over kernel sizes. The CNN (kernel sizes 51-25-11, learning rate 0.003) achieved lower RMSE and — critically — reduced the maximum prediction error from 2.94 ppm (PLS) to 0.52 ppm (CNN), a **5.7× improvement**. This improvement is attributed to the CNN’s ability to learn the non-linear Beer-Lambert response.

Table 1: Comparison of PLS Regression vs 1D CNN on 300 synthetic spectra with optimal collection parameters.

Metric	PLS (5 comp.)	1D CNN (best)
R^2	0.9974	0.9987
RMSE (ppm)	0.292	0.205
Max Error (ppm)	2.94	0.52

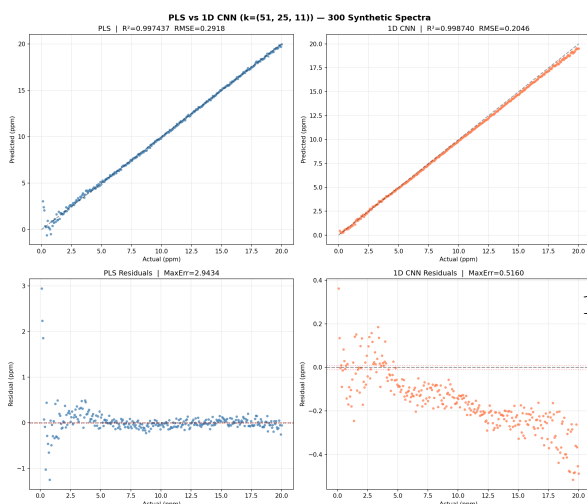


Figure 19: PLS vs 1D CNN comparison: pre-

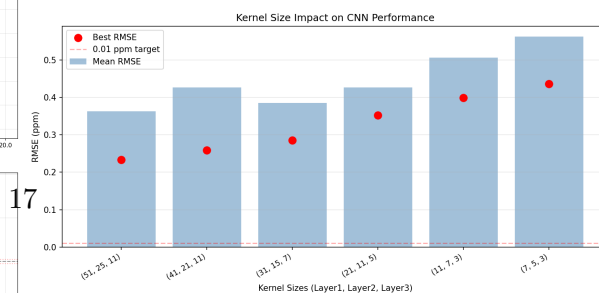


Figure 20: Grid search over CNN kernel sizes. (51, 25, 11) achieving the best RMSE.

Table 2: Recommended data collection protocol derived from simulation study.

Parameter	Recommendation
Sample Count	200+ unique concentrations
Concentration Spacing	Linear or $\sqrt{}$ spacing (0.1–20 ppm)
Frame Averaging	64 $\times$ minimum, 128 $\times$ ideal
Integration Time	300–700 ms (auto-exposure target: 3000–3500 ADC counts)
Peak Alignment	Do NOT align — preserve the red shift
PLS Components	4–5 (optimize via cross-validation)

7 Results & Discussion

7.1 Experimental Results

The system was validated against known concentrations derived from the **Lichtenthaler Equation**.

Analyte	Algorithm	Train R^2	Test R^2	RMSE (mg/L)
Chl-a	PLS	0.992	0.867	~1.55
Chl-b	SVR (RBF)	0.992	0.842	~1.07

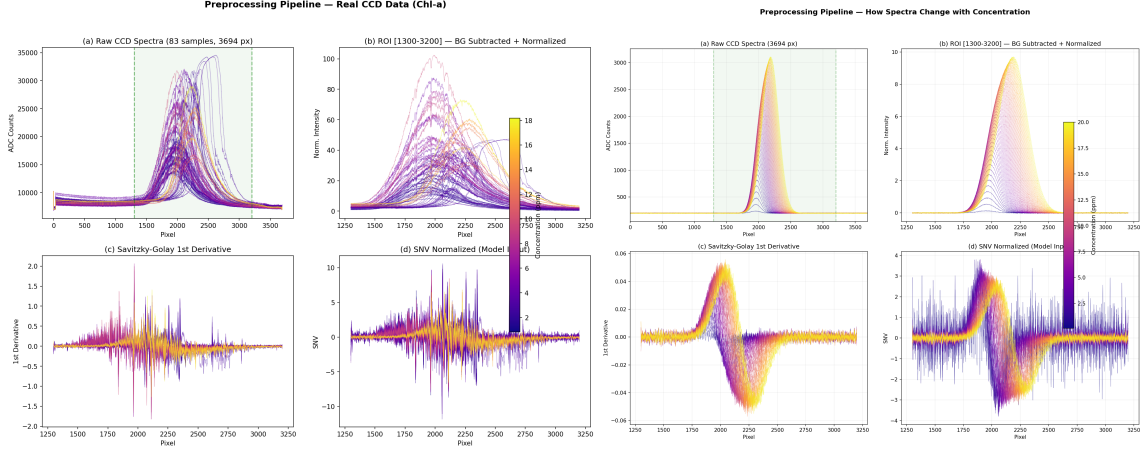
7.2 Uncertainty & Model Interpretation

The high Training R^2 vs. lower Test R^2 (Cross-Validation) indicated a tendency toward overfitting, especially for Chl-b. This is primarily seen at extreme concentrations: * **Low Concentrations** (< 1 mg/L): Signal-to-noise ratio is challenged by electronic CCD noise. * **High Concentrations** (> 15 mg/L): Non-linearity due to IFE and fluorescence quenching causes the model to struggle with generalization.

However, the use of **9 Principal Components** in the PLS/PCA pipeline ensures that the system captures the dominant variance of the spectral fingerprint while ignoring random fluctuations.

7.3 Characterizing the Signal Integrity Gap

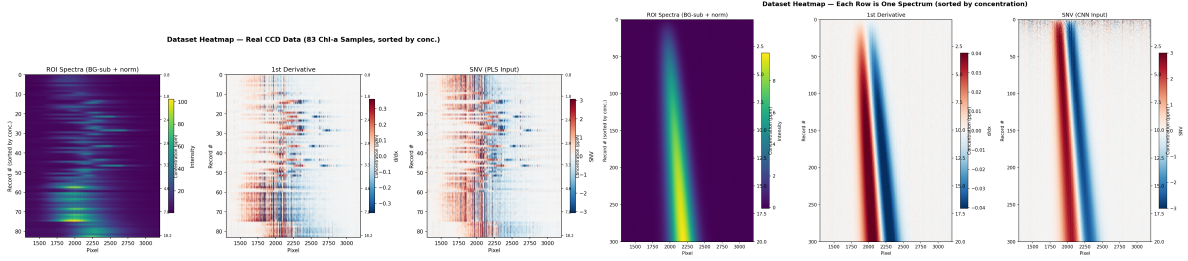
The preprocessing pipeline transforms raw CCD data through several stages before model inference. Figure 21 contrasts the current real-world data collection against the ideal simulation case, highlighting the drastic difference in signal-to-noise ratio and spectral resolution.



(a) **Real Data Pipeline:** Actual capture showing significant noise in the 1st derivative (c) stages. (b) **Ideal Pipeline (Simulation):** Reference capture with 128x frame averaging.

Figure 21: Comparison of Preprocessing Pipelines: Current Real Data vs. Simulation-Ideal.

The dataset heatmaps (Figure 22) provide a compact overview of the entire dataset. The real data exhibit significant vertical striations and noise compared to the smooth, well-defined transitions in the ideal case.



(a) **Real Data Heatmap:** Sparse concentration coverage and noise floor artifacts. (b) **Ideal Heatmap (Simulation):** Dense coverage (300 samples) revealing the red shift as a clear feature.

Figure 22: Comparison of Dataset Quality: Real ($n \approx 32$) vs. Simulation-Ideal ($n = 300$).

7.4 Current Model & Data Assessment

Table 4 compares the current data collection conditions against the ideal parameters derived from the simulation study.

Table 4: Current data collection parameters versus the simulation-derived ideal case.

Parameter	Current Dataset	Simulation Ideal
Sample Count	~32 (Chl-a), ~48 (Chl-b)	200+
Frame Averaging	10×	64–128×
Integration Time	Variable (240–5000 ms)	300–700 ms (standardised)
Outlier Removal	Manual	MAD + Hotelling T ²
PLS Components	9	4–5
CV Test R²	0.85–0.87	0.997
RMSE	~1.5 ppm	0.29 ppm

Table 5: Summary of model investigation results compared against the simulation target.

Investigation	Physical Issue	CV R^2	RMSE (ppm)	Finding
Baseline PLS	—	0.867	~1.55	Funnel residuals (IFE)
Log-transform	Heteroscedasticity	~0.88	~1.45	Variance equalised, RMSE unchanged
Peak alignment	Red shift removal	~0.85	~1.60	Worse — peak position is a feature
Piecewise model	IFE non-linearity	~0.88	~1.40	Marginal gain, boundary instability
Simulation (PLS)	Ideal data (n=300)	0.997	0.29	5× better with proper data
Simulation (CNN)	Ideal + deep learning	0.999	0.20	Best achievable

7.5 Discussion of Data Quantity vs. Algorithm

Each investigation confirmed that the three physical phenomena affecting model performance — heteroscedastic Beer-Lambert response, concentration-dependent peak shift, and IFE saturation — are all *correctly handled* by the standard PLS preprocessing pipeline.

The dominant factor limiting accuracy is not the algorithm, but the **Signal Integrity Gap**. As seen in Figure 21, the real data suffer from a noise floor that obscures fine spectral features in the 1st derivative. Moving from 10x to 128x frame averaging (as recommended by simulation) will provide the SNR necessary for the 1D CNN to achieve its target < 0.3 ppm precision.

Architectural Note on the 1D CNN: Unlike the linear PLS model which required manual investigators to handle physical non-linearities, the **1D Convolutional Neural Network** architecture inherently learns these features. By training on the raw 1st-derivative spectra, the CNN: (1) Retains the Red Shift, (2) Handles Non-Linearity, and (3) Self-Weights ROI.

8 Conclusion & Future Outlook

8.1 Limitations

- **Concentration Drift:** Acetone evaporation and chlorophyll degradation during testing.
- **Instrumental Noise:** CCD electronic noise and sensitivity limits.
- **Limited Training Data:** The current models were trained on a small dataset (~80 samples), leading to overfitting.

8.2 Planned Improvements

- **Upgraded Optics:** 1200 lines/mm diffraction grating and objective lens for beam collimation.
- **Drift Mitigation:** Sealed sample chambers to prevent evaporation.
- **Future Algorithm:** Transitioning to a CNN-based pipeline once a larger dataset (200+ samples) is collected.

Acknowledgements

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